

Transcriptome Demo

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Load required packages (you might have to figure out how to install some of these first...)

```
library(ballgown)
library(RColorBrewer)
library(genefilter)
library(dplyr)
library(devtools)
```

What is this code doing? it creating a data frame called `pheno_data` that stores sample metadata (also known as phenotype data), which will be used to associate each sample with its biological condition (planktonic or biofilm). This is important for differential expression analysis in transcriptomics.

```
pheno_data<-data.frame(ids = c("plank01", "plank02", "biofilm01", "biofilm02"),
                        stage = c("planktonic", "planktonic", "biofilm", "biofilm"))
```

create Ballgown object and check transcript number

```
load("bg.RData")
bg
```

```
## ballgown instance with 5737 transcripts and 4 samples
```

<what is this code doing?> This is a common filtering step before running differential expression, helping to eliminate uninformative or low-variance transcripts that could bias results.

```
bg_filt = subset(bg, "rowVars(texpr(bg)) >1", genomesubset=TRUE)
bg_filt
```

```
## ballgown instance with 5163 transcripts and 4 samples
```

create a table of transcripts

```
results_transcripts<- stattest(bg_filt, feature = "transcript", covariate = "stage",
getFC = TRUE, meas = "FPKM")
results_transcripts<-data.frame(geneNames=geneNames(bg_filt),
transcriptNames=transcriptNames(bg_filt), results_transcripts)
```

choose a transcript to examine more closely (this is a demo, you need to choose another)

```
results_transcripts[results_transcripts$transcriptNames == "gene-PA0002", ]
```

```
## [1] geneNames      transcriptNames feature      id
## [5] fc                pval              qval
## <0 rows> (or 0-length row.names)
```

what information are you given about this transcript? identifying which transcripts show statistically significant changes in expression between biofilm and planktonic conditions. The output from `dim(sigdiff)` tells you how many such transcripts were found.

```
sigdiff <- results_transcripts %>% filter(pval<0.05)
dim(sigdiff)
```

```
## [1] 207  7
```

organize the table <by what metrics is the table being organized?> organizing the sigdiff table by p-value first (smallest to largest), and then by absolute fold change (largest to smallest),

```
o = order(sigdiff[, "pval"], -abs(sigdiff[, "fc"]), decreasing=FALSE)
output = sigdiff[o, c("geneNames", "transcriptNames", "id", "fc", "pval", "qval")]
write.table(output, file="SigDiff.txt", sep="\t", row.names=FALSE, quote=FALSE)
head(output)
```

```
##      geneNames transcriptNames  id      fc      pval      qval
## 4091      .      gene-PA3992 4091 9.886091e+01 0.0003032315 0.9471885
## 4958      .      gene-PA4804 4958 3.563696e-04 0.0006661432 0.9471885
## 2745      .      gene-PA2690 2745 5.783390e-02 0.0014192618 0.9471885
## 2896      tpm      gene-PA2832 2896 1.786570e+03 0.0023414834 0.9471885
## 370      .      gene-PA0365  370 3.964652e-07 0.0023906201 0.9471885
## 3129      pelF      gene-PA3059 3129 1.687425e-03 0.0025838457 0.9471885
```

load gene names

```
bg_table = texpr(bg_filt, 'all')
bg_gene_names = unique(bg_table[, 9:10])
```

pull out gene expression data and visualize

```
gene_expression = as.data.frame(gexpr(bg_filt))
head(gene_expression)
```

```
##          FPKM.plank01 FPKM.plank02 FPKM.biofilm01 FPKM.biofilm02
## .              1.198359    0.9103059      2.526183      2.685373
## MSTRG.1        405.892761  400.8589780      232.324417     181.932617
## MSTRG.10        89.649139   78.5762100       35.010487      59.757320
## MSTRG.100      116.443428  106.2109530       92.206810      95.322479
## MSTRG.1000       7.833186   5.5019700       15.717344      42.342495
## MSTRG.1001       6.845010   4.7381980       38.199095      89.078876
```

<what is this code doing? hint:compare the above output of head(gene_expression) to this output> This code is renaming the columns of the gene_expression data frame to match the sample IDs and then displaying the first few rows and dimensions of the modified data.

```
colnames(gene_expression) <- c("plank01", "plank02", "biofilm01", "biofilm02")
head(gene_expression)
```

```
##          plank01    plank02 biofilm01 biofilm02
## .              1.198359    0.9103059      2.526183      2.685373
## MSTRG.1        405.892761  400.8589780      232.324417     181.932617
## MSTRG.10        89.649139   78.5762100       35.010487      59.757320
## MSTRG.100      116.443428  106.2109530       92.206810      95.322479
## MSTRG.1000       7.833186   5.5019700       15.717344      42.342495
## MSTRG.1001       6.845010   4.7381980       38.199095      89.078876
```

```
dim(gene_expression)
```

```
## [1] 4592    4
```

load the transcript to gene table and determine the number of transcripts and unique genes

```
transcript_gene_table = indexes(bg)$t2g
head(transcript_gene_table)
```

```
##   t_id   g_id
## 1     1 MSTRG.1
## 2     2 MSTRG.2
## 3     3 MSTRG.3
```

```
## 4    4 MSTRG.3
## 5    5 MSTRG.4
## 6    6 MSTRG.5
```

```
length(row.names(transcript_gene_table))
```

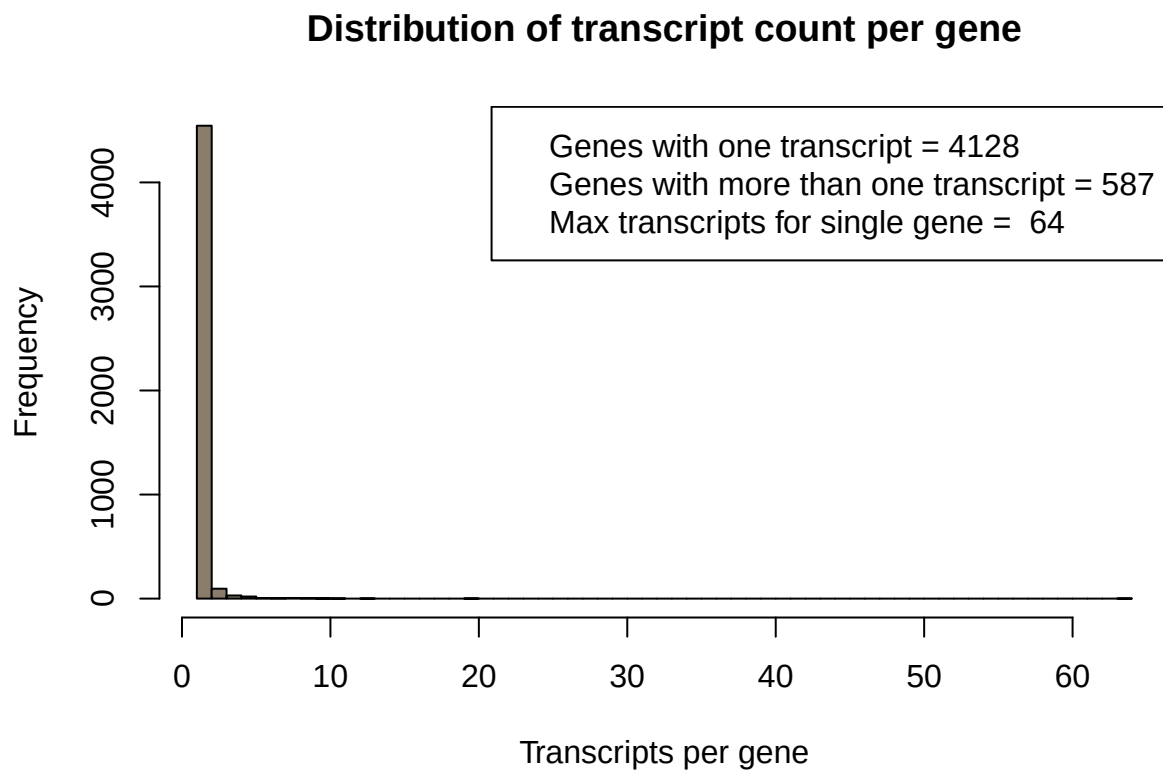
```
## [1] 5737
```

```
length(unique(transcript_gene_table[, "g_id"]))
```

```
## [1] 4715
```

plot the number of transcripts per gene

```
counts=table(transcript_gene_table[, "g_id"])
c_one = length(which(counts == 1))
c_more_than_one = length(which(counts > 1))
c_max = max(counts)
hist(counts, breaks=50, col="bisque4", xlab="Transcripts per gene",
main="Distribution of transcript count per gene")
legend_text = c(paste("Genes with one transcript =", c_one),
paste("Genes with more than one transcript =", c_more_than_one),
paste("Max transcripts for single gene = ", c_max))
legend("topright", legend_text, lty=NULL)
```

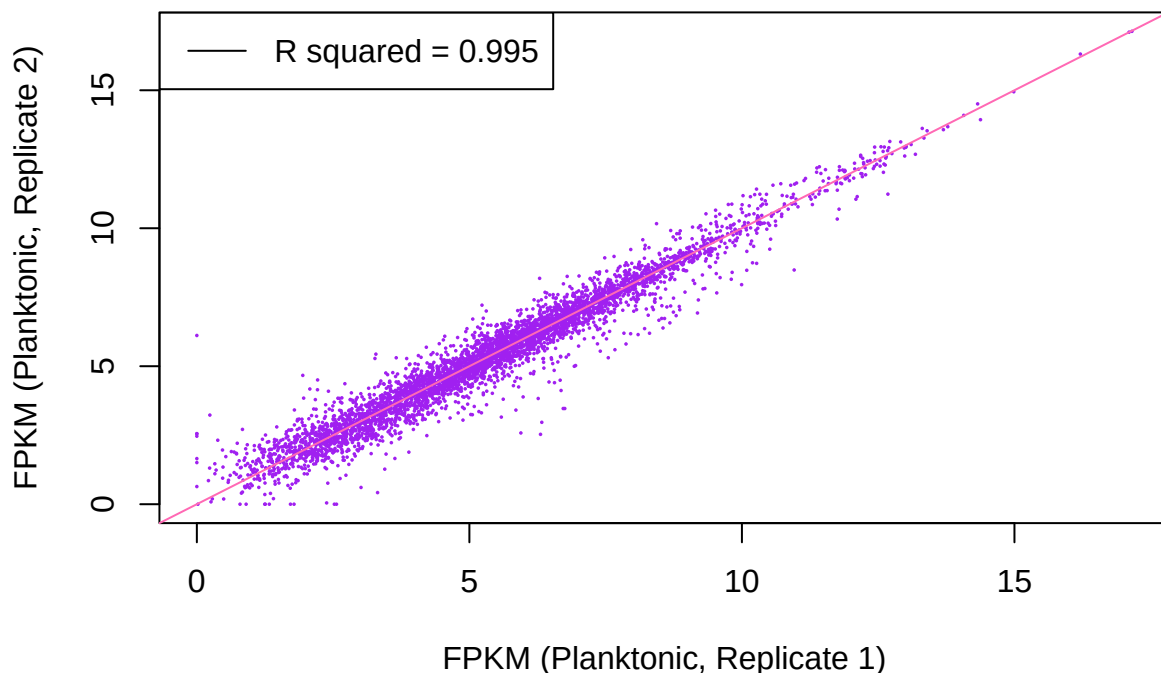


Most of the genes in this transcriptome dataset are simple with the 4128 gene producing only a single transcript. This may suggest limited alternative splicing or transcript variation for the genes in *P. aeruginosa*. Smaller transcript like the 587 genes indicates transcriptional complexity or variation. The 74 distinct transcripts has the highest complexity.

create a plot of how similar the two replicates are for one another. We have two data sets... how can you modify this code in another chunk to create a plot of the other set? By replacing the plank01 and plank02 with biofilm01 and biofilm02.

```
x = gene_expression[, "plank01"]
y = gene_expression[, "plank02"]
min_nonzero=1
plot(x=log2(x+min_nonzero), y=log2(y+min_nonzero), pch=16, col="purple", cex=0.25,
xlab="FPKM (Planktonic, Replicate 1)", ylab="FPKM (Planktonic, Replicate 2)",
main="Comparison of expression values for a pair of replicates")
abline(a=0,b=1, col = "hotpink")
rs=cor(x,y)^2
legend("topleft", paste("R squared = ", round(rs, digits=3), sep=""), lwd=1, col="black")
```

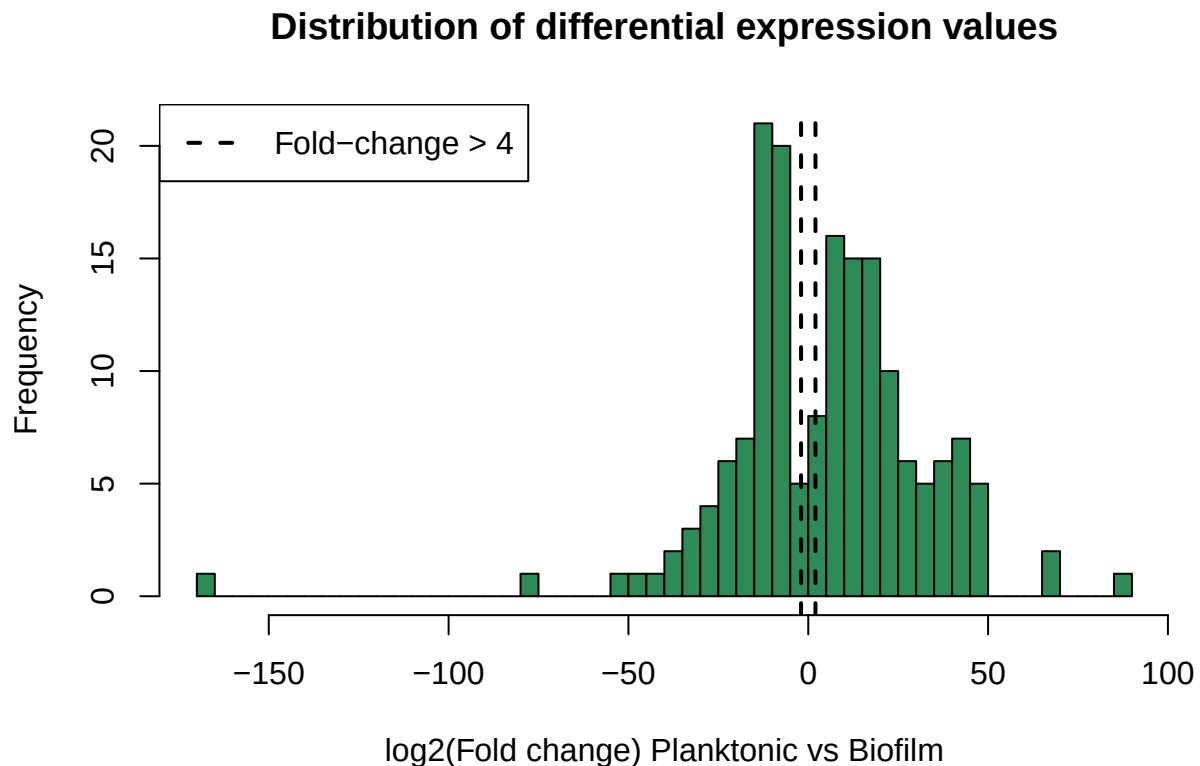
Comparison of expression values for a pair of replicates



What does it mean if the two data sets are similar? It would mean that the biological replicates are consistent and reproducible this means identical gene expression patterns. This shows consistency, minimal error, and good expression.

create plot of differential gene expression between the conditions

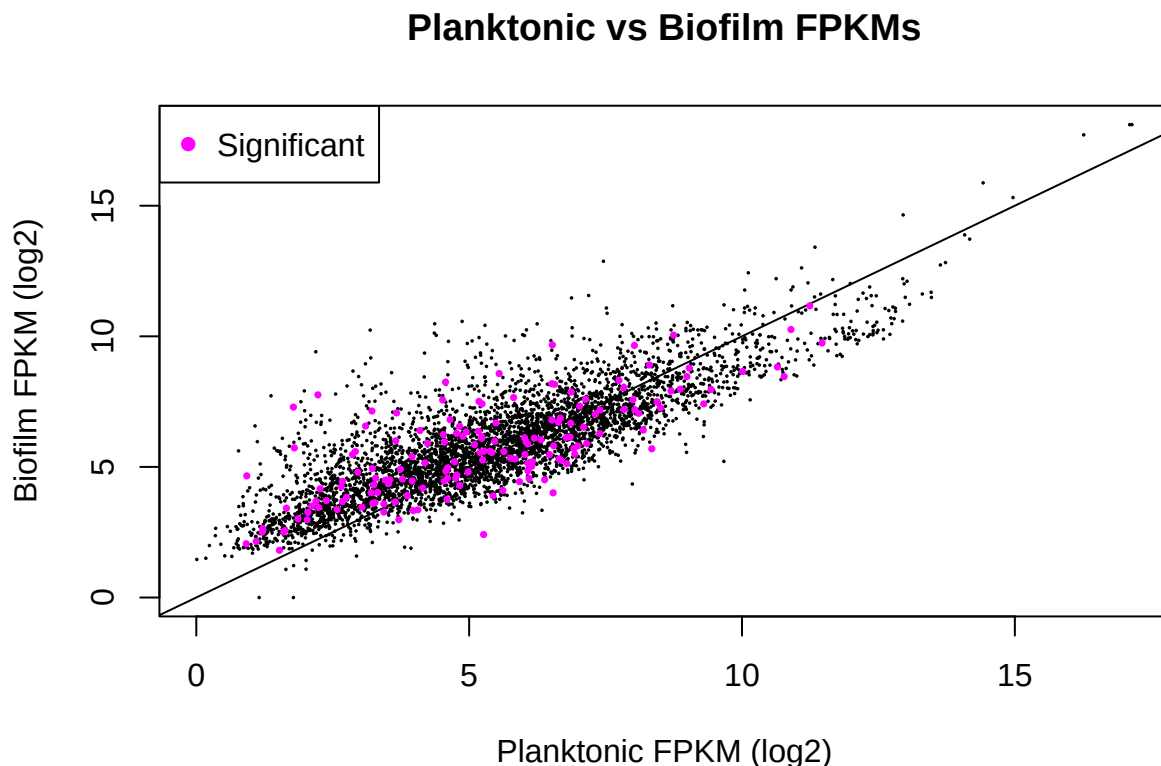
```
results_genes = statstest(bg_filt, feature="gene", covariate="stage", getFC=TRUE, meas="FPKM")
results_genes = merge(results_genes,bg_gene_names,by.x=c("id"),by.y=c("gene_id"))
sig=which(results_genes$pval<0.05)
results_genes[, "de"] = log2(results_genes[, "fc"])
hist(results_genes[sig,"de"], breaks=50, col="seagreen",
xlab="log2(Fold change) Planktonic vs Biofilm",
main="Distribution of differential expression values")
abline(v=-2, col="black", lwd=2, lty=2)
abline(v=2, col="black", lwd=2, lty=2)
legend("topleft", "Fold-change > 4", lwd=2, lty=2)
```



interpret the above figure: This shows frequency over $\log_2(\text{Fold change})$ Plan-
 lktinic vs Biofilm. There is a peak around the negative numbers just before 0
 showing a decrease abundance in biofilm condition with an increase in features
 and the peak drops at and around 0 because both decrease abundance and de-
 crease in features of and this slope grows but then decreases continues decreasing
 going into the positive numbers because of leveling out in data after $\log_2$ grows.

Plot total gene expression highlighting differentially expressed
 genes

```
gene_expression[, "plank"] = apply(gene_expression[, c(1:2)], 1, mean)
gene_expression[, "biofilm"] = apply(gene_expression[, c(3:4)], 1, mean)
x = log2(gene_expression[, "plank"] + min_nonzero)
y = log2(gene_expression[, "biofilm"] + min_nonzero)
plot(x=x, y=y, pch=16, cex=0.25, xlab="Planktonic FPKM (log2)", ylab="Biofilm FPKM (log2)",
     main="Planktonic vs Biofilm FPKMs")
abline(a=0, b=1)
xsig=x[sig]
ysig=y[sig]
points(x=xsig, y=ysig, col="magenta", pch=16, cex=0.5)
legend("topleft", "Significant", col="magenta", pch=16)
```



make a table of FPKM values

```
fpkm = texpr(bg_filt, meas="FPKM")
```

choose a gene to determine individual expression (pick a different number than I did) index 3

```
ballgown::transcriptNames(bg_filt)[3]
```

```
##          3  
## "gene-PA0003"
```

```
ballgown::geneNames(bg_filt)[3]
```

```
##      3  
## "recF"
```

transform to log2

```
transformed_fpkm <- log2(fpkm[3, ] + 1)
```

make sure values are properly coded as numbers

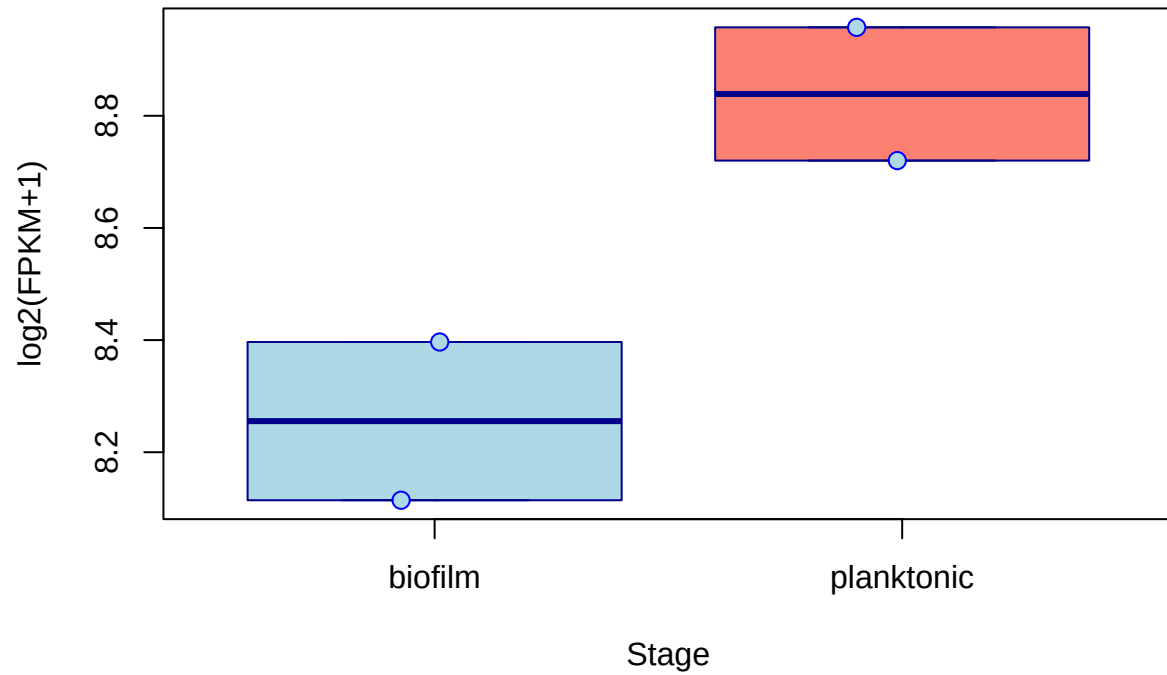
```
numeric_stages <- as.numeric(factor(pheno_data$stage))
```

```
jittered_stages <- jitter(numeric_stages)
```

plot expression of individual gene

```
boxplot(transformed_fpkm ~ pheno_data$stage,  
         main=paste(ballgown::geneNames(bg_filt)[3], ' : ', ballgown::transcriptNames(bg_filt)[3]),  
         xlab="Stage",  
         ylab="log2(FPKM+1)",  
         col=c("lightblue", "salmon"),  
         border="darkblue")  
  
points(transformed_fpkm ~ jittered_stages,  
        pch=21, col="blue", bg="lightblue", cex=1.2)
```

recF : gene-PA0003



interpret the above figure: This figure shows $\log_2(\text{FPKM}+1)$ vs biofilm and planktonic. It shows biofilm values for $\log_2$ from around 8.0-8.4 which represents a slightly lower expression and planktonic values around 8.7-9.0 which shows higher expression of the (3) gene. This all means that the planktonic state is more